Golden Cross & Death Cross EMA Trading Strategy

Group 8, Cohort 1

UCLA Anderson

The Strategy

UCLA Anderson School of Management

Exponential Moving Average (EMA)

- Simple Moving Average (SMA or MA) - **equally weighted** average of N periods (day, 1 hour, 15 min, etc) at closing price
- EMA weighs newest price data more than older price data
- Why is this better for our strategy?
  - **EMA reacts more quickly to recent price changes**
  - **Less lag than SMA**

Golden Cross & Death Cross

- Golden Cross
  - Short-term moving average crosses above long-term moving average
  - Recent prices are higher than long-term average
  - Buying momentum -> BULLISH
- Death Cross
  - Short-term moving average crosses below long-term moving average
  - Recent prices are lower than long-term average
  - Selling momentum -> BEARISH

Strategy

- Technical indicators
  - **Short - period exponential moving average (8 EMA)**
  - **Long - period exponential moving average (25 EMA)**
- "Buy" signal
  - **8EMA crosses above 25EMA (Golden Cross) on daily time frame**
- "Sell" (Close) signal - no shorting
  - **8EMA crosses below 25EMA (Death Cross) on daily time frame**
- Transactions made at closing price

Visualization - S&P 500 (^SPX)

UCLA Anderson

Trading Performance

UCLA Anderson School of Management

Performance Summary

	Strategy	Buy-and-Hold
Annual Return	17%	10%
Annual Volatility	5.49	6.88
Sharpe Ratio	0.68	0.37

- Portfolio (S&P 500 Energy Sector ^GSPE) Beta: -0.008
- Performance Alpha: 0.014

Cumulative Returns Comparison

- Strategy consistently outperformed Buy-and-Hold.
- Higher cumulative return, especially in volatile markets.

UCLA Anderson

Rolling Volatility & Sharpe Ratio

- Volatility was initially very high indicating significant market fluctuations.
- Declined steadily and remained stable around 0.4
- The Sharpe Ratio shows that the strategy has mostly positive risk-adjusted returns.

UCLA Anderson

The Economic Intuition Behind Our Strategy

UCLA Anderson School of Management

Bullish vs Bearish Signals

- **Imperative to find economic signals for buying /selling efficiently:**
  - **Golden Cross** → **Signals <u>Economic Expansion:</u>**
    - Broad Market Optimism

      (Rising investor confidence in sustained growth)
    - Positive Institutional Awareness

      (Reinforcing momentum in an index)

Bullish vs Bearish Signals

- **Imperative to find economic signals for buying /selling efficiently:**
  - **Death Cross** → **Signals <u>Economic Contraction</u>**
    - Recession Fears

      (Investors expect slower economic growth or potential recessions)
    - Institutional Selling Pressure

      (Reinforcing the bearish momentum by reducing equity exposure)

Why Does This Work on an Index?

- **Unlike single stocks, index funds represent the overall market**
  - **Momentum persists in broad indices**
  - **Macro-economic drivers**
  - **Long-term edge**

Ideal Market Conditions

- **Intuitively, what economic (market) conditions work best for our strategy to be successful?**
  - **During strong macroeconomic trends**
  - **In highly liquid environments**
  - **When volatility is low**

Market Frictions

UCLA Anderson School of Management

Factors

1. Liquidity Constraints

- **Impact**: Low-liquidity stocks or ETFs can lead to slippage and higher execution costs.
- **Opportunity**: Trading during high-liquidity periods (such as market opening and closing hours) could help minimize slippage.
- **Adaptation**: Focus on index ETFs or liquid derivatives that reflect broad market movements to avoid thinly traded assets.

2. Delayed Reaction from Institutional Investors

- **Impact**: Institutional investors often adjust positions gradually to avoid market impact.
- **Opportunity**: Retail traders can enter trades faster and exit before large institutions fully adjust.
- **Adaptation**: Monitoring unusual volume spikes in ETFs and sector-specific indices may help anticipate institutional movements.

Factors

3. Order Flow Imbalance

- **Impact**: Large buy/sell orders from institutional players create temporary price dislocations.
- **Opportunity**: EMA crossovers often align with institutional accumulation or distribution phases.
- **Adaptation**: Pair EMA signals with volume analysis to confirm trend strength.

4. Behavioral Market Frictions

- **Impact**: Retail investors often panic-sell or chase price momentum inefficiently.
- **Opportunity**: The Golden Cross can capitalize on bullish sentiment before broader participation.
- **Adaptation**: Implement sentiment indicators (e.g., Put-Call ratios, Fear & Greed Index) to validate signals.

Factors

5. Market Microstructure Inefficiencies

- **Impact**: Bid-ask spreads widen in volatile conditions, affecting trade execution.
- **Opportunity**: Using limit orders instead of market orders can reduce transaction costs.
- **Adaptation**: Timing trades when spreads are narrow (e.g., during market stability) can improve execution.

6. Trend-Persistent Market Environments

- **Impact**: Trending markets sustain directional movement, making EMA strategies effective.
- **Opportunity**: Identifying macroeconomic conditions (GDP growth, interest rate trends) helps confirm momentum-based strategies.
- **Adaptation**: Avoid choppy, sideways markets where false crossovers may occur.

Key Risk Factors & Limitations

UCLA Anderson School of Management

Key Risk Factors & Limitations

- Lagging Indicator:
  - **Signals based on past data can cause you to enter or exit trades too late**
- Whipsaw Signals:
  - **Rapid market fluctuations can trigger false signals, leading to a costly series of trades**
- Delayed Entry and Exit:
  - **The strategy's confirmation delay means you miss optimal price points, reducing gains.**

Key Risk Factors & Limitations

- Lack of Fundamental Analysis:
  - **Ignoring fundamentals leaves you vulnerable to unexpected events that technical signals can't predict.**
- Market Regime Dependence:
  - **The strategy thrives only in trending markets and struggles during choppy, sideways conditions.**
- Execution and Slippage & No Transaction Cost Considered:
  - **Real-world trading costs and imperfect execution can substantially diminish the strategy's theoretical returns.**

...

Thank you!